



Universidad Autónoma De Tamaulipas

Fundamentos de Programación

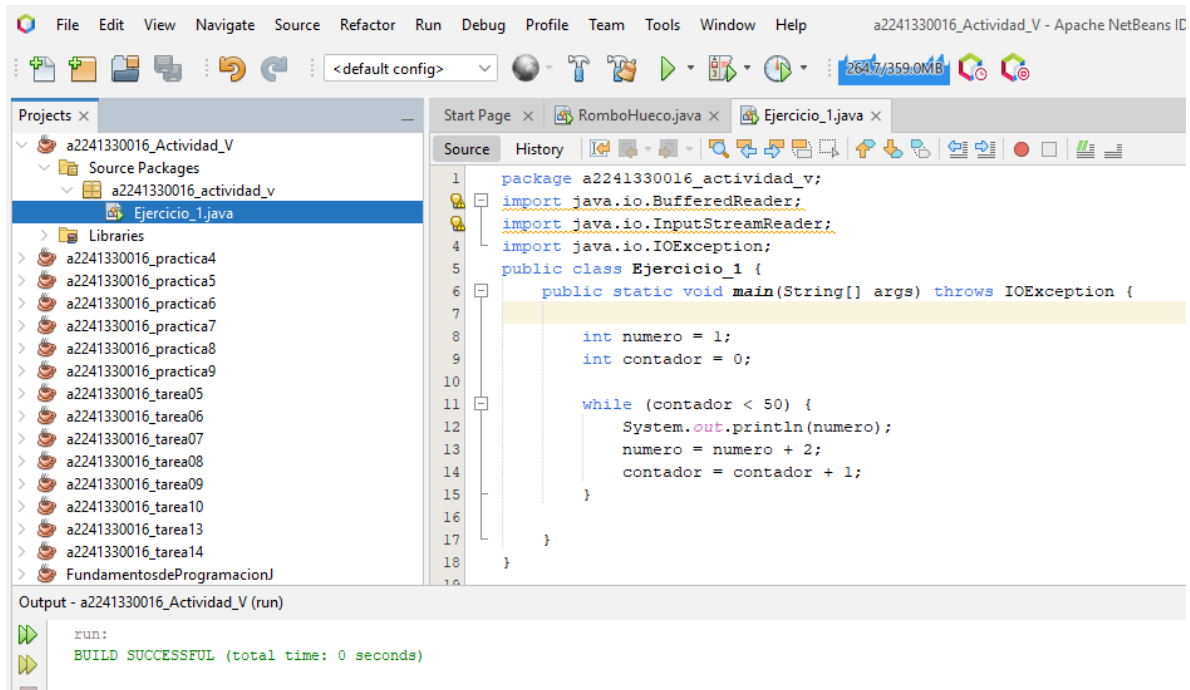
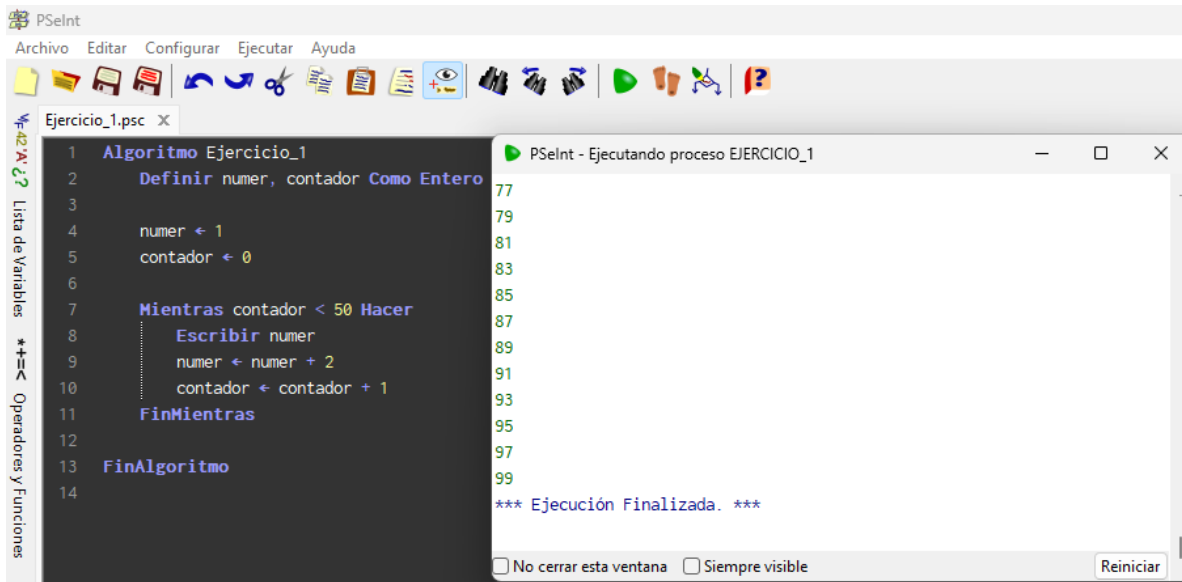
1-N

Actividad en clase

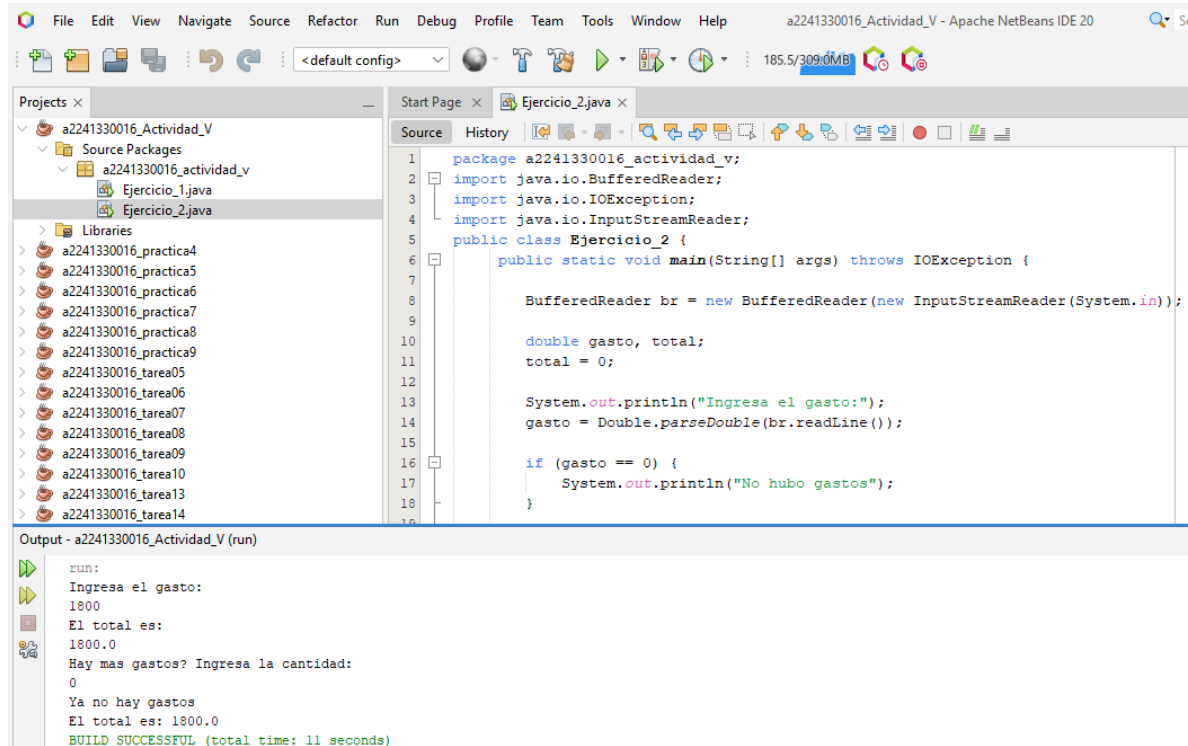
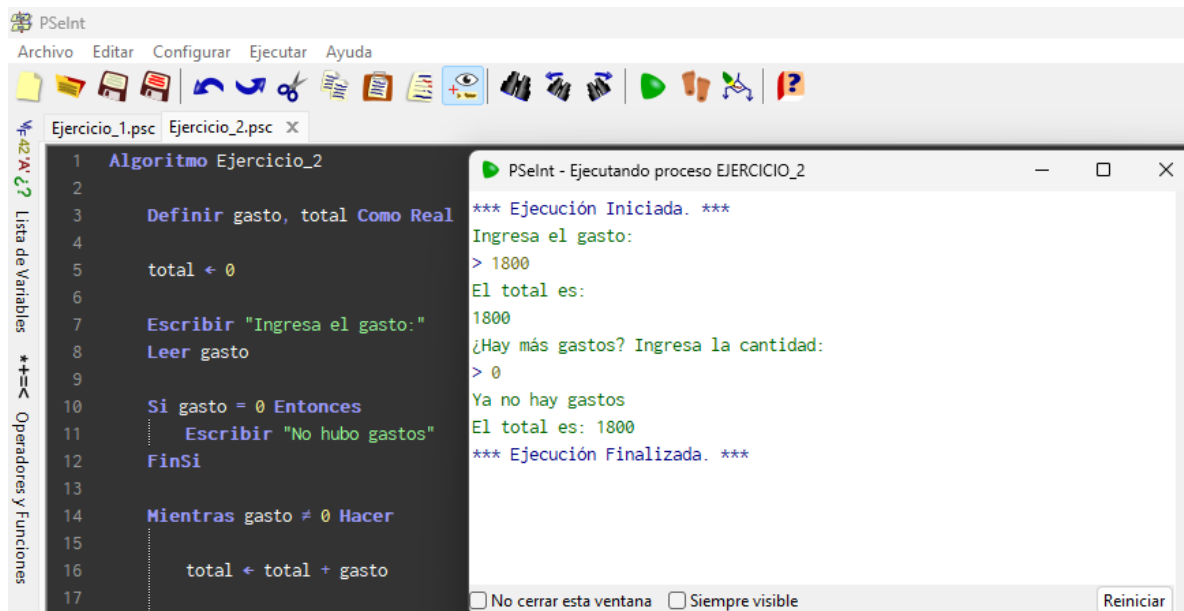
Molina Meneses Diego

While

Ejercicio1:



Ejercicio2:



Ejercicio3:

The screenshot shows the PSeInt IDE with a pseudocode program named 'Proceso Ejercicio_3'. The program defines a variable 'numero' and 'cubo' as Real, prompts the user to enter a natural number, and calculates its cube. It includes a conditional check to ensure the input is a natural number (greater than 0 and less than or equal to 9). The execution window on the right shows the program's output, including the user's input and the calculated cube.

```

1  Proceso Ejercicio_3
2
3  Definir numero, cubo Como Real
4
5  numero ← 0
6  cubo ← 0
7
8  Escribir "Ingresa el número natural"
9  Leer numero
10
11 Si numero < 0 O numero > 9 Entonces
12     Escribir "El número no es natural"
13 FinSi
14
15 Mientras numero ≥ 0 Y numero ≤ 9 Hacer
16     cubo ← numero * 3
17

```

Execution Output:

```

343
Ingresa otro número natural
> 7
El cubo del número es:
343
Ingresa otro número natural
> 7
El cubo del número es:
343
Ingresa otro número natural
> 11
El número no es natural
*** Ejecución Finalizada. ***

```

The screenshot shows the Apache NetBeans IDE with a Java program named 'Ejercicio_3.java'. The program uses Java's standard input/output classes to read a number from the user and calculate its cube. It includes a conditional check to ensure the input is a natural number. The output window at the bottom shows the program's execution results, including the user's input and the calculated cube.

```

1 package a2241330016_actividad_v;
2 import java.io.BufferedReader;
3 import java.io.IOException;
4 import java.io.InputStreamReader;
5 public class Ejercicio_3 {
6     public static void main(String[] args) throws IOException {
7
8         BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
9
10        double numero = 0;
11        double cubo = 0;
12
13        System.out.println("Ingresa el numero natural:");
14        numero = Double.parseDouble(br.readLine());
15
16        if (numero < 0 || numero > 9) {
17            System.out.println("El numero no es natural");
18        }
19    }
20 }

```

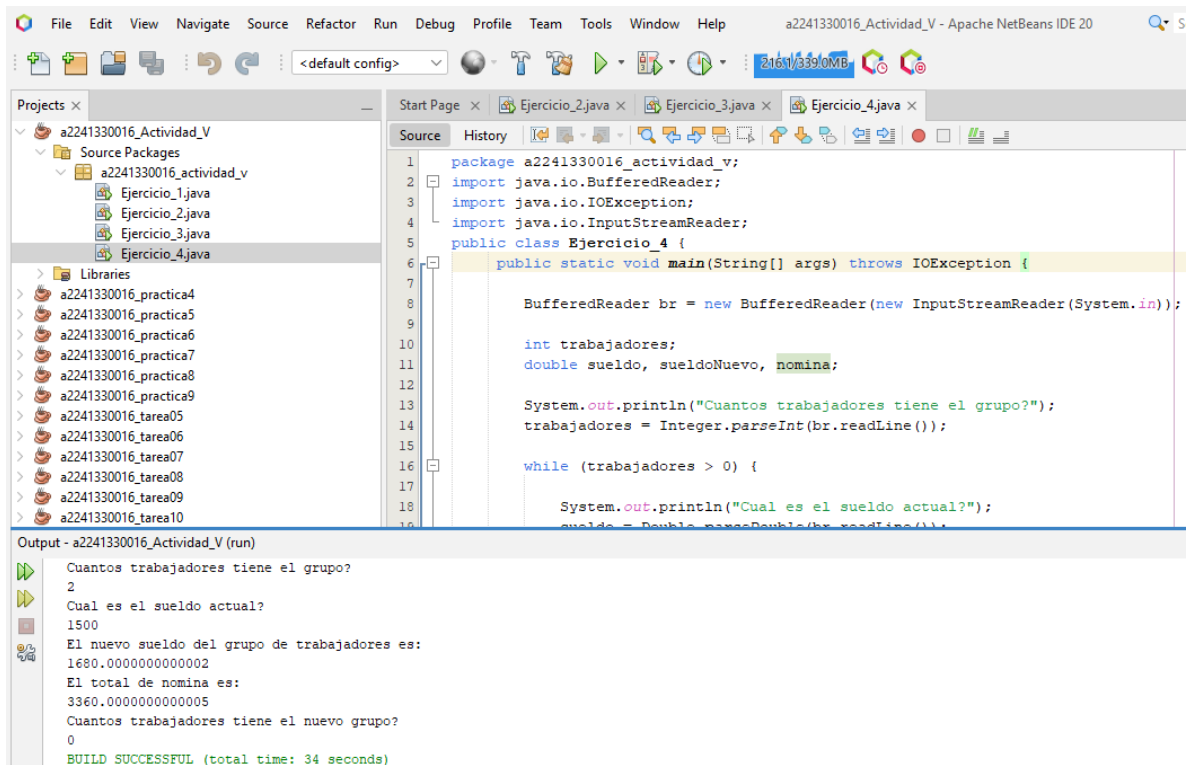
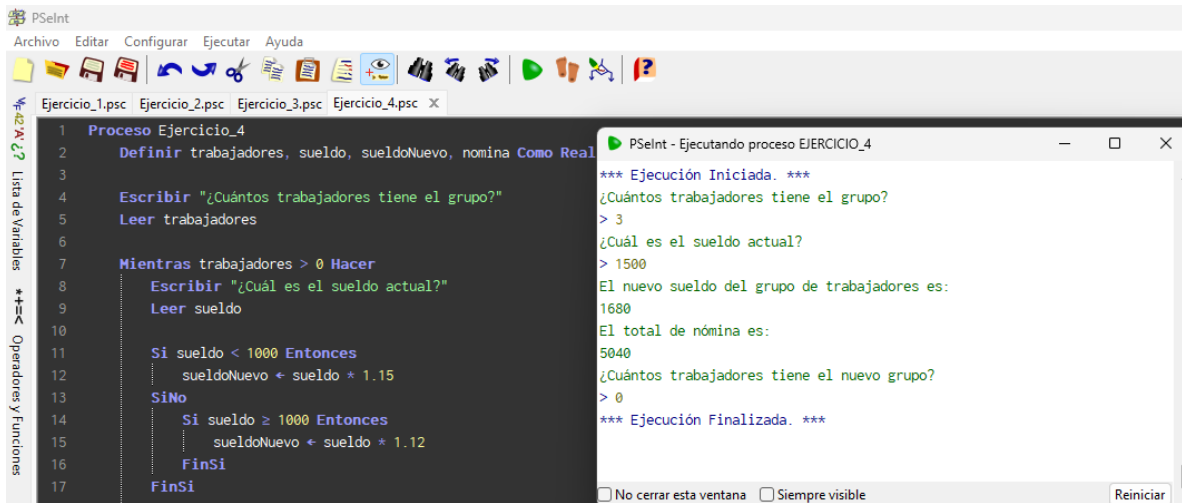
Output - a2241330016_Actividad_V (run)

```

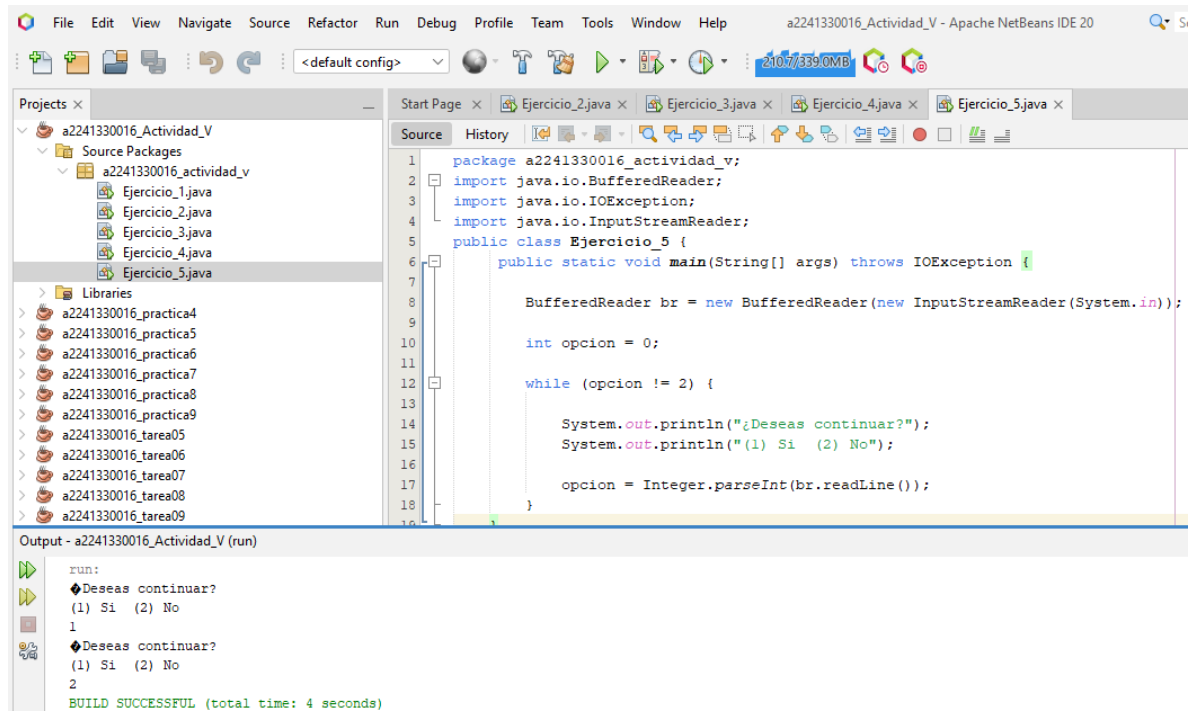
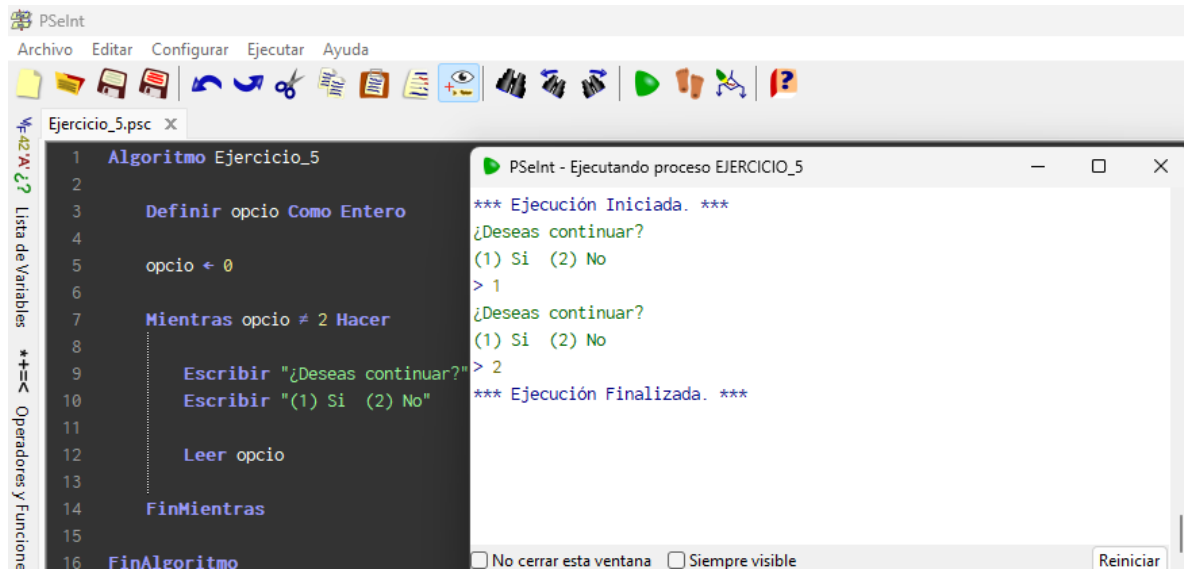
4
El cubo del numero es:
64.0
Ingresa otro numero natural:
6
El cubo del numero es:
216.0
Ingresa otro numero natural:
11
El numero no es natural
BUILD SUCCESSFUL (total time: 14 seconds)

```

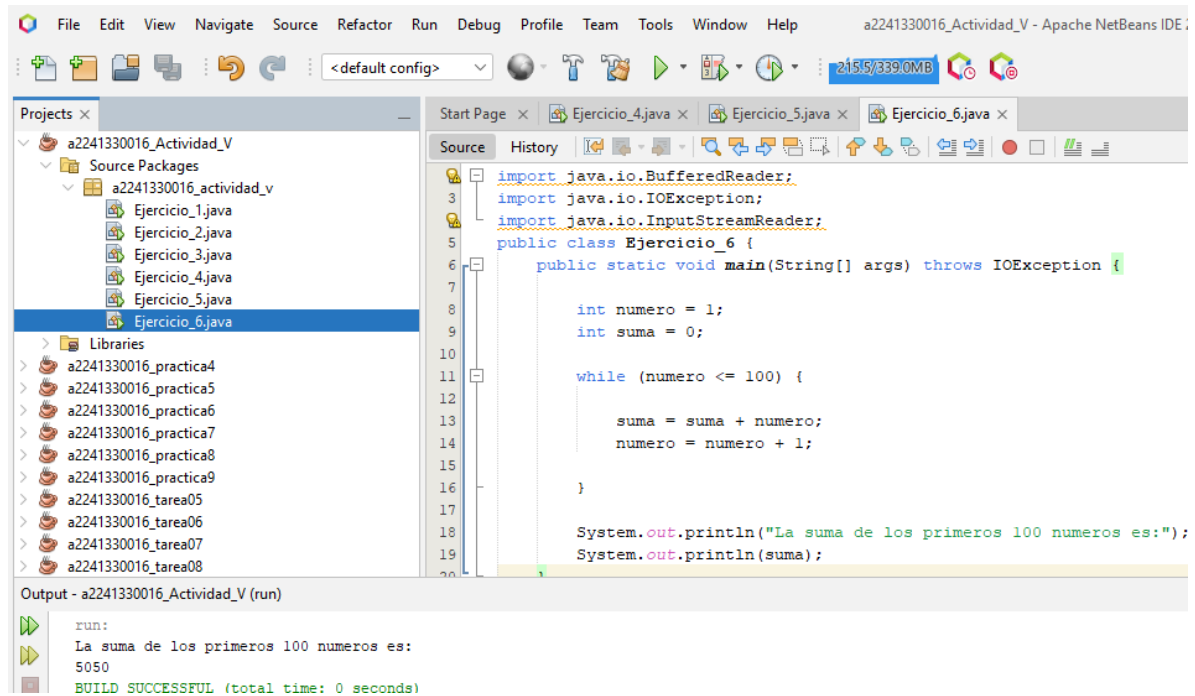
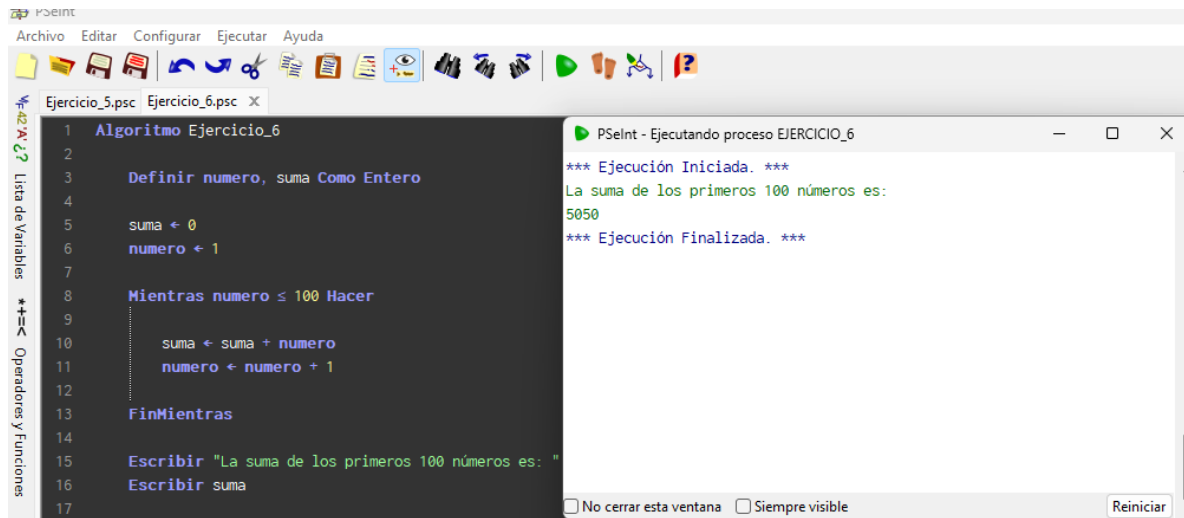
Ejercicio4:



Ejercicio5:



Ejercicio6:



Ejercicio7:

PSelnt

Archivo Editar Configurar Ejecutar Ayuda

Ejercicio_5.psc Ejercicio_6.psc Ejercicio_7.psc X

```
1 Algoritmo Ejercicio_7
2   Definir numero Como Entero
3
4   numero ← 1
5
6   Mientras numero ≤ 1000 Hacer
7
8       Escribir numero
9       numero ← numero + 1
10
11   FinMientras
12 FinAlgoritmo
13
```

Lista de Variables

Operadores y Funciones

PSelnt - Ejecutando proceso EJERCICIO_7

```
989
990
991
992
993
994
995
996
997
998
999
1000
*** Ejecución Finalizada. ***
```

☐ No cerrar esta ventana ☐ Siempre visible Reiniciar

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help a2241330016_Actividad_V - Apache NetB

<default config>

233.5/339.0MB

Projects X

- a2241330016_Actividad_V
 - Source Packages
 - a2241330016_actividad_v
 - Ejercicio_1.java
 - Ejercicio_2.java
 - Ejercicio_3.java
 - Ejercicio_4.java
 - Ejercicio_5.java
 - Ejercicio_6.java
 - Ejercicio_7.java
 - Libraries
 - a2241330016_practica4
 - a2241330016_practica5
 - a2241330016_practica6
 - a2241330016_practica7
 - a2241330016_practica8
 - a2241330016_practica9
 - a2241330016_tarea05
 - a2241330016_tarea06
 - a2241330016_tarea07

Start Page X Ejercicio_6.java X Ejercicio_7.java X

Source History

```
1 package a2241330016_actividad_v;
2 import java.io.BufferedReader;
3 import java.io.IOException;
4 import java.io.InputStreamReader;
5 public class Ejercicio_7 {
6     public static void main(String[] args) throws IOException {
7
8         int numero = 1;
9
10        while (numero <= 1000) {
11
12            System.out.println(numero);
13            numero = numero + 1;
14        }
15    }
16 }
```

Output - a2241330016_Actividad_V (run)

```
991
992
993
994
995
996
997
998
999
1000
BUILD SUCCESSFUL (total time: 0 seconds)
```

Ejercicio8:

The screenshot shows the PSeInt IDE with a Pascal program named 'Ejercicio_8.psc'. The program calculates the sum of a series starting at 0 and increasing by 2 until it reaches 1800. The output window shows the sequence of numbers from 1777 to 1800, the total sum 648720, and a confirmation that the execution is finished.

```
1 Algoritmo Ejercicio_8
2
3 Definir serie, contador Como Entero
4
5 serie ← 0
6 contador ← 0
7
8 Escribir "A continuación, se imprime"
9
10 Mientras serie < 1800 Hacer
11     serie ← serie + 2
12     contador ← contador + serie
13     Escribir serie
14
15     serie ← serie + 3
16     contador ← contador + serie
```

Output:

```
1777
1780
1782
1785
1787
1790
1792
1795
1797
1800
La suma de la serie es:
648720
*** Ejecución Finalizada. ***
```

The screenshot shows the Apache NetBeans IDE with a Java program named 'Ejercicio_8.java'. The program calculates the sum of a series starting at 0 and increasing by 2 until it reaches 1800. The output window shows the sequence of numbers from 1782 to 1800, the total sum 648720, and a confirmation that the build was successful.

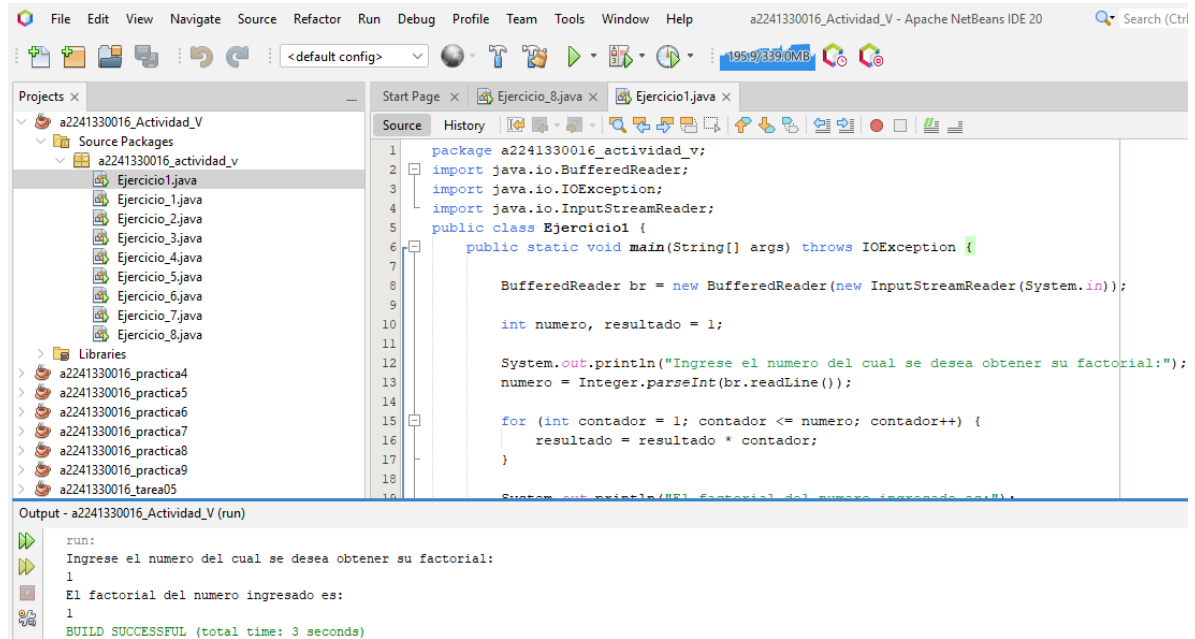
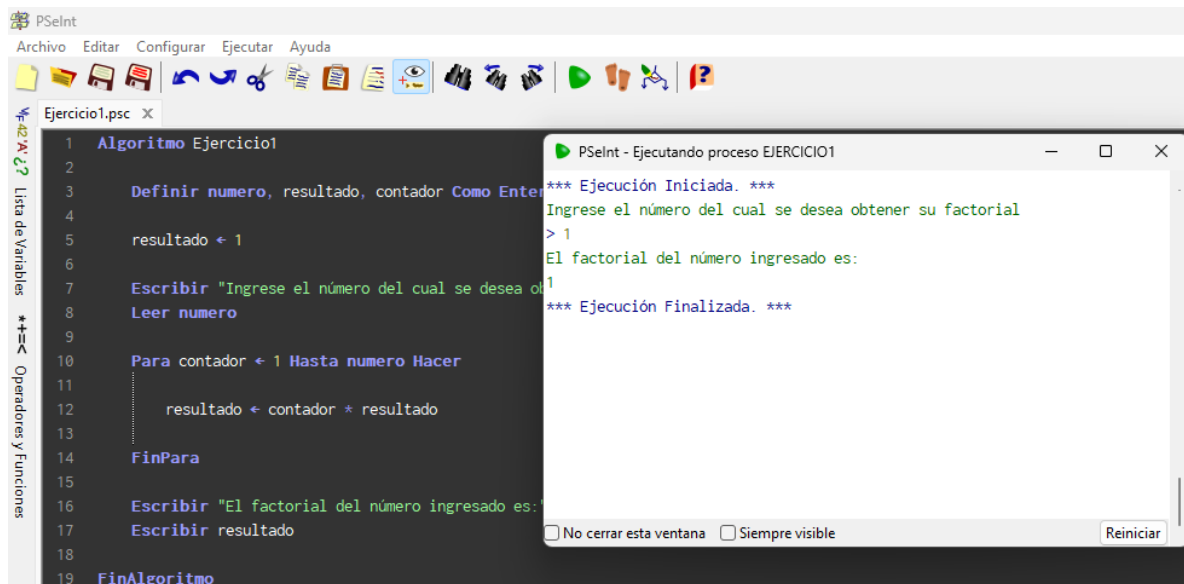
```
1 package a2241330016_actividad_v;
2 import java.io.BufferedReader;
3 import java.io.IOException;
4 import java.io.InputStreamReader;
5 public class Ejercicio_8 {
6     public static void main(String[] args) throws IOException {
7
8         int serie = 0;
9         int contador = 0;
10
11         System.out.println("A continuación, se imprime una serie de numeros.");
12
13         while (serie < 1800) {
14
15             serie = serie + 2;
16             contador = contador + serie;
17             System.out.println(serie);
18
19             serie = serie + 3;
20         }
21     }
22 }
```

Output - a2241330016_Actividad_V (run)

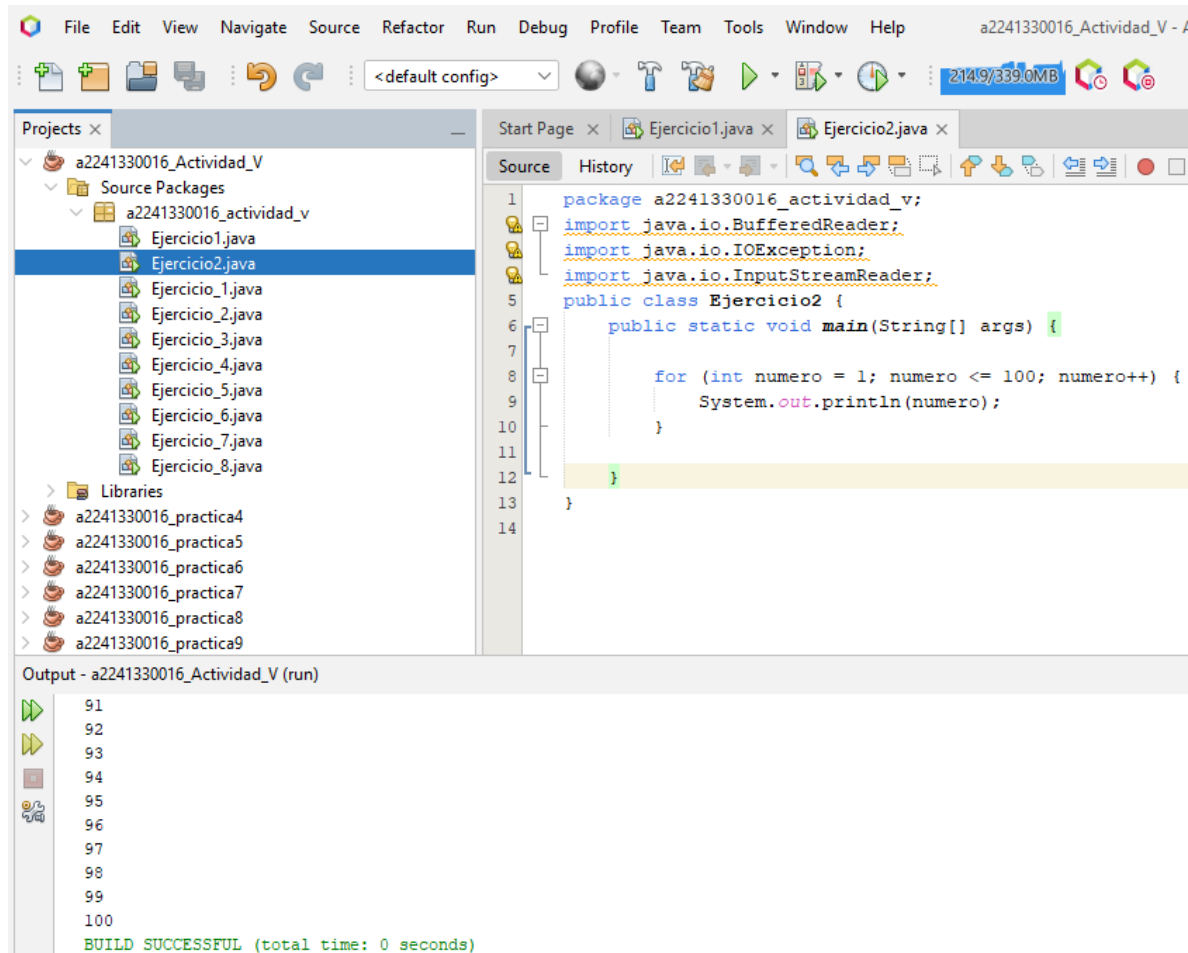
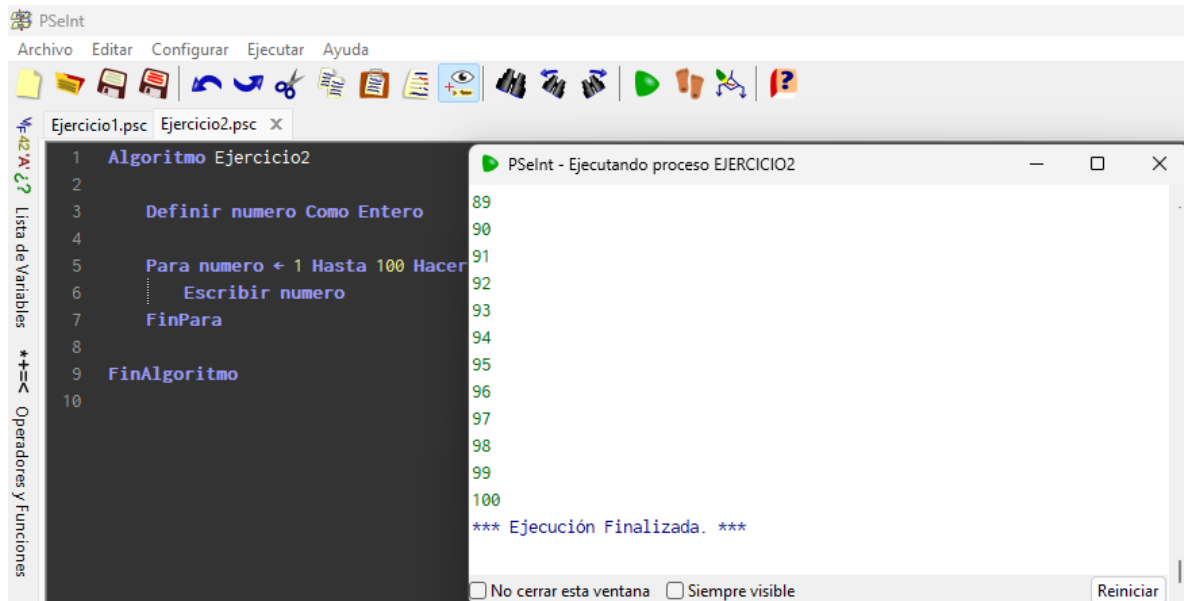
```
1782
1785
1787
1790
1792
1795
1797
1800
La suma de la serie es:
648720
BUILD SUCCESSFUL (total time: 0 seconds)
```

For

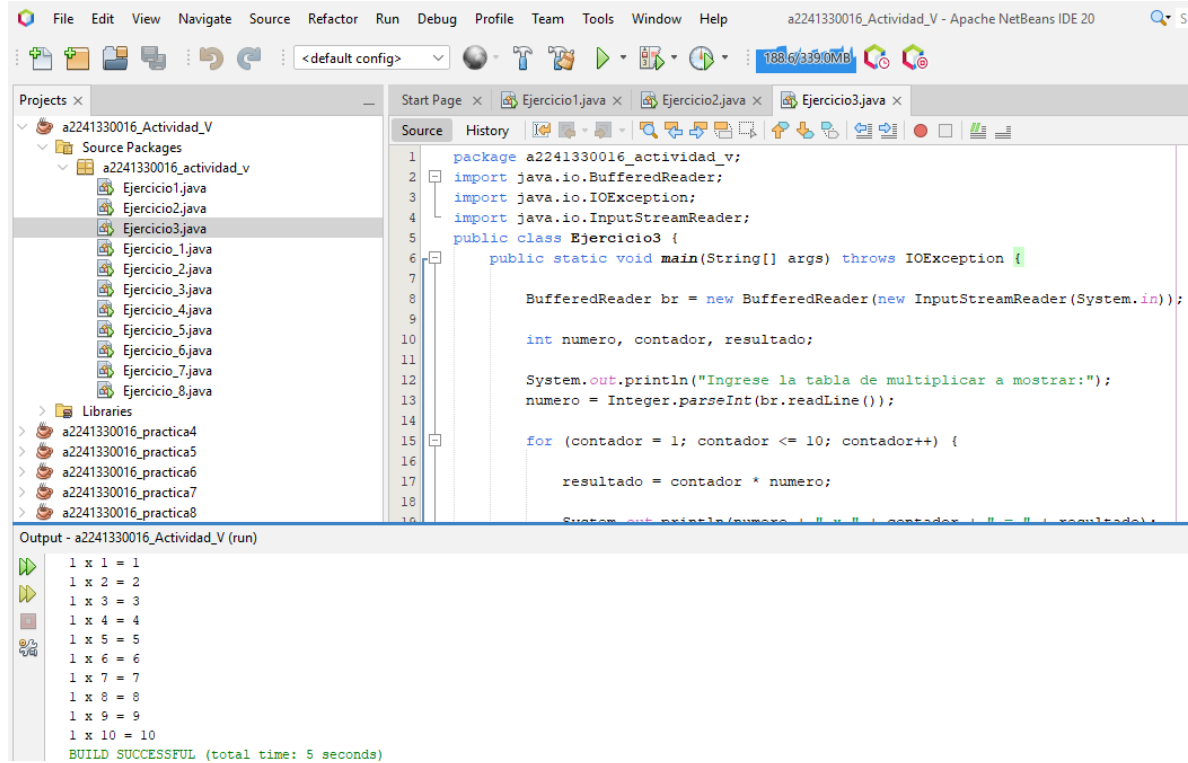
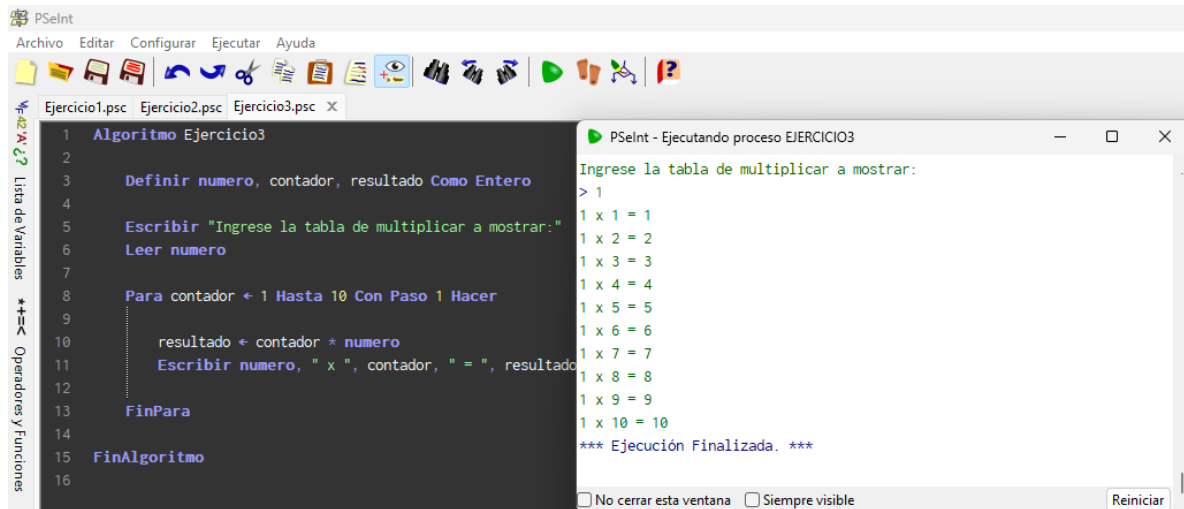
Ejercicio1:



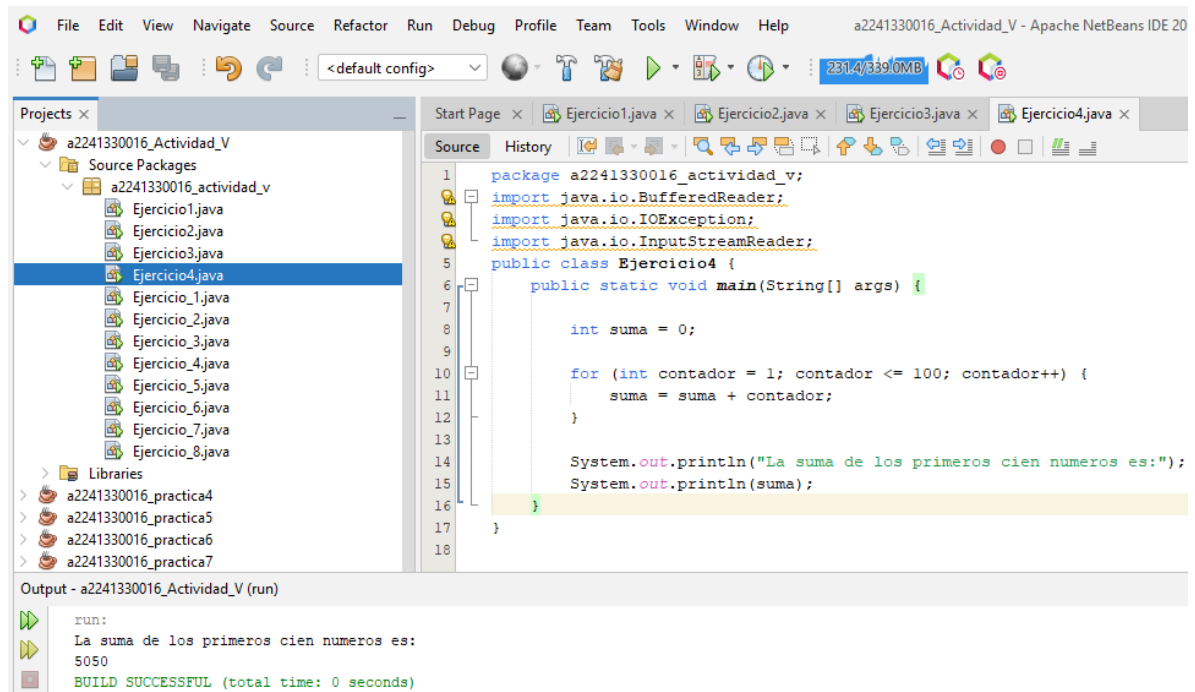
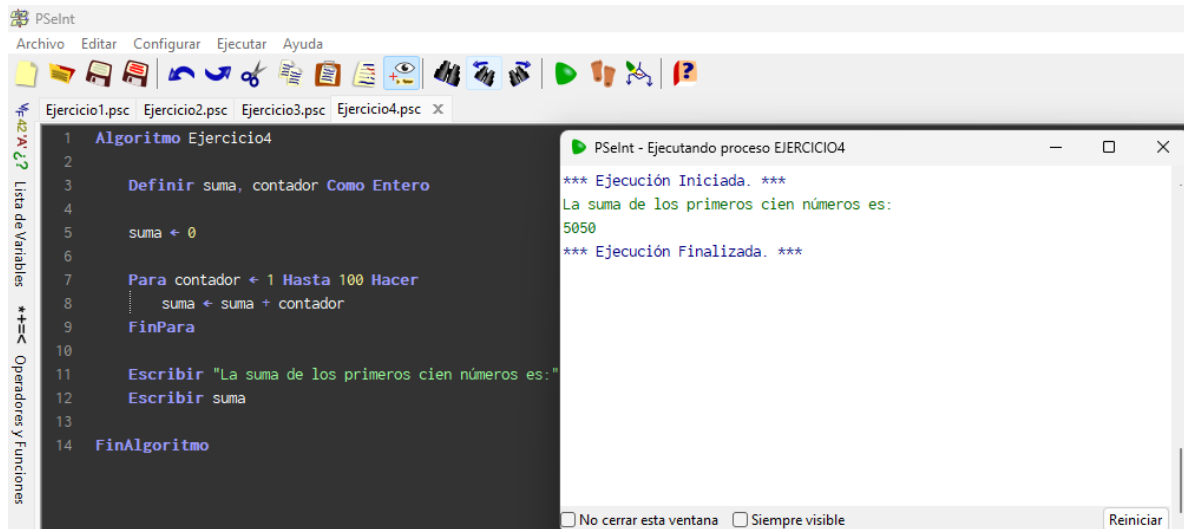
Ejercicio2:



Ejercicio3:



Ejercicio4:



Ejercicio5:

The screenshot shows the PSeInt IDE interface. On the left, a list of variables and operators is visible. The main editor displays a pseudocode algorithm for 'Ejercicio5'. The algorithm defines variables 'contador', 'resultado', and 'tabla' as integers. It uses a 'Para' loop to iterate 'tabla' from 1 to 9, and an inner 'Para' loop to iterate 'contador' from 1 to 10. Inside the inner loop, it calculates 'resultado = contador * tabla' and prints the expression 'tabla x contador = resultado'. The execution output on the right shows the resulting multiplication table from 8 x 9 to 9 x 10, followed by a message indicating the execution is finished.

```
1 Algoritmo Ejercicio5
2   Definir contador, resultado, tabla Como Entero
3
4   Para tabla ← 1 Hasta 9 Con Paso 1 Hacer
5
6       Para contador ← 1 Hasta 10 Con Paso 1 Hacer
7
8           resultado ← contador * tabla
9           Escribir tabla, " x ", contador, " = ", resultado
10
11       FinPara
12
13   FinPara
14 FinAlgoritmo
15
```

8 x 9 = 72
8 x 10 = 80
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
*** Ejecución Finalizada. ***

The screenshot shows the Apache NetBeans IDE. The 'Projects' pane on the left shows a project named 'a2241330016_Actividad_V' with a sub-project 'a2241330016_actividad_v' containing several Java files. The 'Source' pane shows the code for 'Ejercicio5.java'. The code imports 'BufferedReader', 'IOException', and 'InputStreamReader', and defines a 'main' method that uses nested 'for' loops to calculate and print the multiplication table. The 'Output' pane at the bottom shows the execution results, which match the output from the PSeInt IDE, followed by a 'BUILD SUCCESSFUL' message.

```
1 package a2241330016_actividad_v;
2 import java.io.BufferedReader;
3 import java.io.IOException;
4 import java.io.InputStreamReader;
5 public class Ejercicio5 {
6     public static void main(String[] args) {
7
8         int contador, resultado, tabla;
9
10        for (tabla = 1; tabla <= 9; tabla++) {
11
12            for (contador = 1; contador <= 10; contador++) {
13
14                resultado = contador * tabla;
15                System.out.println(tabla + " x " + contador + " = " + resultado);
16            }
17        }
18    }
19 }
```

9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
BUILD SUCCESSFUL (total time: 0 seconds)

Para Si

Ejercicio1:

